

Real Analysis

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Table of contents

Preface

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1 Introduction

This is a book created from markdown and executable code.

See Knuth (1984) for additional discussion of literate programming.

Theorem 1.1 (Fermat's First Theorem). *This extension has been written by a teacher who hopes that students read the course notes...*

Please look back to Theorem ??

Lemma 1.2. *This extension has been written by a teacher who hopes that students read the course notes...*

Exercise 1.3. This extension has been written by a teacher who hopes that students read the course notes...

Example 1.4. This extension has been written by a teacher who hopes that students read the course notes...

1.1 Acknowledgements

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Part I

The Real Numbers

As discussed in the introduction, our approach to real analysis will be to start by laying out a set of axioms that we *assume* are true about the real number system $\mathbb{R}$ and then *prove* everything else based upon this set of axioms.

In order to properly take this approach, we need a set of axioms that:

- contains enough information to let us prove everything there is to know about $\mathbb{R}$ (all of calculus, for example),
- is specific enough so that it distinguishes $\mathbb{R}$ from other number systems, and
- is as simple as possible

The purpose of the first bullet point is clear enough: if I don't assume anything, I won't be able to prove anything.

The second bullet point about distinguishing $\mathbb{R}$ from other number systems is related, but deserves some explanation. Consider the following proposition:

For every $x > 0$, there exists y such that $y^2 = x$.

This is true if our universe is $\mathbb{R}$ but false if our universe is $\mathbb{Q}$. If our axioms for $\mathbb{R}$ do not distinguish it from $\mathbb{Q}$, we will have no hope of proving such a proposition. In short, I want a set of axioms that is true for $\mathbb{R}$ *but for no other number system*.

Note

I am using the term *number system* a bit loosely. For our purposes, though, it's fine to think of any set with binary operations corresponding to addition and multiplication—think $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.

Notes

- Should include a thought experiment about how do we know there is a number x such that $e^x + x = 1$ (with Desmos graphic).
- Should include something about how mathematics shifted from a search for absolute truth to a search for understanding of how consequences follow from hypotheses.

2 The Field and Order Axioms

2.1 Highlights

- Two of the three axioms we will assume about the real number system: the field axiom which gives us the rules of arithmetic, and the order axiom which gives us the rules for inequalities.
- We see examples of other number systems besides $\mathbb{R}$ that also satisfy one or both of these axioms. In particular $\mathbb{Q}$ also satisfies both.

2.2 Preparation

Look ahead to the definition of a field (Definition ??) and definition of an ordered field (Definition ??). See if you can convince yourself the real number system does indeed satisfy both of these definitions. For example, every real number x *does* in fact have an additive inverse (a number $-x$ that be added to x to obtain 0); it *is* true that when you multiply two positive real numbers you get another positive real number.

Take another moment to scan some of the *theorems* that follow each of these axioms. None of these should be surprising—they are all well-known facts about how arithmetic and inequalities work in $\mathbb{R}$ —but what may be surprising is that extremely simple facts like $0x = 0$ (Theorem ??) and $1 > 0$ (Theorem ??) can be *proven* from even simpler assumptions.

For now, let's focus on thinking about which other familiar number systems satisfy the field and order axioms.

Exercise 2.1. Think about some familiar number systems that *aren't* the real numbers: $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{C}$. You may rely on prior knowledge for this problem—no need to show all of your work.

1. Which of these are fields? For those that are not, which properties do they fail to have?
2. Which of these are ordered fields? Which of these satisfy the two conditions of Definition ?? despite not actually being a field?

3. **(Exploratory)** You should have found that one of the four number systems is a field and an ordered field. The Intermediate Value Theorem (for example) *is* true in $\mathbb{R}$, but it is *not* true in any of these other four number systems. With that information, do you think one would be able to prove the Intermediate Value Theorem in $\mathbb{R}$ if our *only* assumptions about $\mathbb{R}$ are the field and order axioms?

2.3 Fields

For our purposes, a *number system* $\mathbb{S}$ is a set with two binary operations $+$ and $\cdot$. This is a nonstandard term, but we won't use it much beyond this section.

Definition 2.2. A number system $\mathbb{S}$ is said to be a *field* if it satisfies the following properties.

1. Addition is commutative (for all $a, b \in \mathbb{S}$, $a + b = b + a$).
2. Addition is associative (for all $a, b, c \in \mathbb{S}$, $(a + b) + c = a + (b + c)$).
3. There is an element $0 \in \mathbb{S}$, called the *additive identity* such that for all $a \in \mathbb{S}$, $a + 0 = a$.
4. Every $a \in \mathbb{S}$ has an *additive inverse* $-a$ with the property that $a + (-a) = 0$.
5. Multiplication is commutative (for all $a, b \in \mathbb{S}$, $a \cdot b = b \cdot a$).
6. Multiplication is associative (for all $a, b, c \in \mathbb{S}$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$).
7. There is an element $1 \in \mathbb{S}$ different from 0, called the *multiplicative identity* such that for all $a \in \mathbb{S}$, $a \cdot 1 = a$.
8. Every $a \in \mathbb{S} \setminus \{0\}$ has an *additive inverse* a^{-1} with the property that $a + (a^{-1}) = 1$.
9. The *distributive rule* holds (for all $a, b, c \in \mathbb{S}$, $a(b + c) = a \cdot b + a \cdot c$).

Collectively, these properties tell us how **arithmetic** works for fields. And the first axiom we make about the real number system is that it is indeed a field.

Theorem 2.3. $\mathbb{R}$ is a field.

Note that there is no reference to the operations of subtraction and division in the definition of a field, but these can be *defined* in terms of the established properties: $a - b$ is defined to be $a + (-b)$ (after all, $-b$ is defined above), and if $b \neq 0$, we define $\frac{a}{b}$ to be $a \cdot b^{-1}$ (after all, b^{-1} is defined above).

What is somewhat remarkable is that *all* of the other rules of arithmetic for the real numbers can be derived from the field axiom. For example, you may have supposed that the following theorem would be an assumption, but one can prove it from the even more basic assumptions in the definition of a field.

Theorem 2.4. For any $x \in \mathbb{R}$, $0x = 0$.

Exercise 2.5. Prove Theorem ??.

 Hint

Note that Theorem ?? involves both multiplication and, implicitly, addition (since it involves the *additive* identity). Thus its proof must involve the one part of the field axiom that combines addition and multiplication: the distributive property.

(More theorems with the caveat that we won't keep doing this for long.)

Any facts that we can prove about $\mathbb{R}$ that use *only* the assumption that $\mathbb{R}$ is a field, will in fact be true for *any* field. And there are some weird fields.

Exercise 2.6. Consider the number system $\mathbb{F}_2 = \{\alpha, \beta\}$ consisting of just two elements, where the addition and multiplication operations are defined by

$$\alpha + \alpha = \beta \quad \alpha + \beta = \alpha \quad \beta + \alpha = \alpha \quad \beta + \beta = \beta$$

and

$$\alpha \cdot \alpha = \alpha \quad \alpha \cdot \beta = \beta \quad \beta \cdot \alpha = \beta \quad \beta \cdot \beta = \beta.$$

It turns out that $\mathbb{F}_2$ is a field. Which element of $\mathbb{F}_2$ is 0? Which is 1? Which is -1 ?

 Hint

Asking which element is 0 means asking which element is the additive identity—the thing that can be added to any element and get that same element back. Similarly for the multiplicative identity 1. And asking which element is -1 means asking which element is *the additive inverse of 1* (the thing you add to 1 and get back 0).

So the field axiom alone fails miserably at the goal of distinguishing $\mathbb{R}$ from other number systems—it cannot even distinguish $\mathbb{R}$ from a number system with just two elements! So we need more axioms capturing more of the structure of $\mathbb{R}$.

2.4 Order and Inequalities

Another important feature of the real number system is the behavior of inequalities and how they interact with arithmetic. This is captured by the next axiom.

Definition 2.7. A field $\mathbb{F}$ is called an *ordered field* if it has a relation $>$ with the following properties.

1. For all $a \in \mathbb{F}$, exactly one of the following is true:

- $a > 0$,
- $a = 0$, or
- $-a > 0$.

2. If $a > 0$ and $b > 0$, then $a + b > 0$ and $a \cdot b > 0$.
3. $b > a$ if and only if $b - a > 0$.

We can then *define* relations $<, \leq, \geq$ in the usual way based on the relation $>$ ¹.

Theorem 2.8. $\mathbb{R}$ is an ordered field.

Again, the properties listed in Definition ?? are not surprising; we already knew they were true for $\mathbb{R}$. But more surprising is the fact the everything else we know about how inequalities behave and interact with arithmetic can be prove from Theorem ?? and Theorem ??.

Theorem 2.9. $1 > 0$.

You are probably familiar with the basics of how inequalities and arithmetic interact, and henceforth, we can use them at will without citation. In this class, being fluent with the rules for inequalities will be essential, though, so we collect a list of “inequality rules” (all of which could in principle be proven from Theorem ?? and Theorem ??) in Section ??.

Worth reflecting on now is where we stand on our stated goals in creating the axioms. The Theorem ?? was successful in distinguishing $\mathbb{R}$ from $\mathbb{F}_2$ because there is no ordering on $\mathbb{F}_2$ satisfying Definition ?? (you will prove this in Exercise ?? below).

However, we can note that $\mathbb{Q}$ is also an ordered field since it satisfies Theorem ??. This means that anything we can prove directly from Theorem ?? and Theorem ?? will apply just as well if our universe were $\mathbb{Q}$ as it would if our universe were $\mathbb{R}$. In particular, we would have no hope of proving, say, the Intermediate Value Theorem, which is true in $\mathbb{R}$ but not in $\mathbb{Q}$.

This all means that we need another assumption—an axiom that will distinguish the real number system from the rationals in a way that will allow us to prove everything else we’ll need in order to do calculus.

Exercise 2.10. Prove Theorem ?? using only the Field and Order Axioms.

Exercise 2.11. Prove that the following fields are *not* ordered.

1. $\mathbb{F}_2$ (defined in Exercise ??)
2. The complex numbers $\mathbb{C}$

¹That is, we define $a < b$ to mean $b > a$, $a \leq b$ to mean $a < b$ or $a = b$, and $a \geq b$ to mean $a > b$ or $a = b$.

 Hint

This is a good opportunity to use a proof by contradiction. What contradiction can you create by assuming there *is* an ordering on $\mathbb{F}_2$ satisfying the conditions of Definition ??? What about in $\mathbb{C}$?

The following exercise says something very simple, but ends up being very useful in practice.

Exercise 2.12. Suppose $r \geq 0$ and that for all $\epsilon > 0$, $r \leq \epsilon$. Prove that $r = 0$.

2.4.1 Review Questions

1. Give some examples of number systems that are fields. Give some examples of number systems that are not fields.
2. Give some examples of fields that are ordered fields. Give some examples of fields that are not ordered fields.
3. Summarize for yourself how we can be certain that it is impossible to prove a statement like “every positive number in $\mathbb{R}$ has a square root” building off of only the field and order axioms.

2.5 Inequality Rules

- a is positive $\Leftrightarrow -a$ is negative
- $1 > 0$
- $x \leq y$ and $y \leq z \Rightarrow x \leq z$
- $a > b$ and $c \geq d \Rightarrow a + c > b + d$
- $a > b > 0$ and $c \geq d > 0 \Rightarrow ac > bd$
- $a > b$ and $c < 0 \Rightarrow ac < bc$
- $a < 0$ and $b < 0 \Rightarrow a + b < 0$
- $a < 0$ and $b < 0 \Rightarrow ab > 0$
- $a > 0$ and $b < 0 \Rightarrow ab < 0$
- $a > 0 \Leftrightarrow \frac{1}{a} > 0$
- $a > 1 \Rightarrow a^2 > a$
- $0 < a < 1 \Rightarrow a^2 < a$
- $a > b > 0 \Rightarrow a^2 > b^2$

3 The Least Upper Bound Axiom

3.1 Highlights

- The definition of the supremum of a set and the Least Upper Bound axiom of $\mathbb{R}$
- The Archimedean Properties of $\mathbb{R}$

3.2 Preparation

Leading up to the third and final axiom of $\mathbb{R}$, we make a few definitions.

Definition 3.1. Let $S \subseteq \mathbb{R}$ be a subset of the real numbers.

1. We say $b \in \mathbb{R}$ is an *upper bound* of S if for all $x \in S$, $b \geq x$. If S has an upper bound, we say S is *bounded above*.
2. We say that $u \in \mathbb{R}$ the *least upper bound* or *supremum* of S if for all upper bounds b of S , $u \leq b$, and in this case we write

$$u = \sup S.$$

3. If u is the least upper bound S and $u \in S$, we say that u is the *greatest element* of S .

Exercise 3.2.

1. Write down three upper bounds for the interval $S = [0, 3)$. Does S have a least upper bound; if so what is it? Does S have a greatest element; if so, what is it?
2. For each set below, say what its supremum and greatest element are or say why it doesn't have one.
 - a. $\{\frac{n}{n+1} : n \in \mathbb{N}\}$
 - b. $(2, \infty)$
 - c. $\{\frac{1}{n} : n \in \mathbb{N}\}$
 - d. $(-\infty, 3]$
 - e. $\{x : x^2 \leq 3\}$

3. Describe in your own words which subsets of $\mathbb{R}$ have upper bounds, which have least upper bounds, and which have greatest elements. This doesn't have to be precise; just give me the vibe.
4. **(Exploratory)** How would you change your answers to the previous problems if our universe were $\mathbb{Q}$ instead of $\mathbb{R}$?

Exercise 3.3. Write “ S is bounded above” in logical symbolic notation; see if you can do so without using the term “upper bound”.

Exercise 3.4. Suppose b is an upper bound on a set S . What does it mean for b to *not* be the *least* upper bound of S ?

3.3 Least Upper Bounds

Part 2e of Exercise ?? is meant to draw a contrast between how least upper bounds work in $\mathbb{R}$ versus how they work in $\mathbb{Q}$. In particular, the problem asks, if

$$S = \{x : x^2 \leq 3\}?$$

does S have a least upper bound?

Whether our universe is $\mathbb{Q}$ or $\mathbb{R}$ or some other ordered field, S certainly has *an* upper bound; $b = 2$, for example, is greater than any element of S (that is, 2 is greater than any number whose square is less than 3). But as you may have guessed, an ordered field has a number whose square is 3 if and only if the set S defined above has a *least* upper bound (in which case $\sup S$ is such a number).

We have a sense that the since 3 has no square root in $\mathbb{Q}$, there is something missing—a “hole” in the rationals that we plug by instead thinking about $\mathbb{R}$. And as this example shows, plugging such a hole can be related to the existence of a least upper bound on a set.

This connection goes in the opposite direction as well. If we *assume* the supremum of S exists, then we at least have a candidate for a number whose square is 3 (actually proving this is more complicated than you might expect—see Theorem ?? below).

This is all motivation for the final axiom we'll assume about the real numbers: a guarantee of the existence of least upper bounds that will have the effect of ensuring that any $\mathbb{R}$ will not have any “holes” in this sense, the way that $\mathbb{Q}$ does.

Theorem 3.5 (The Least Upper Bound Axiom). *Every non-empty subset of $\mathbb{R}$ that is bounded above has a least upper bound in $\mathbb{R}$.*

 Tip

Note that using Theorem ?? to conclude that a subset $S \subseteq \mathbb{R}$ has a supremum requires S to satisfy two hypotheses:

- S is nonempty, and
- S is bounded above.

From now on, we'll make a point of emphasizing which theorems that require the Least Upper Bound Axiom in their proofs and how. Theorems that rely on the Least Upper Bound Axiom will be highlighted with a maroon bar.

As a first step to understanding least upper bounds, we should justify the use of the word “the” in Definition ??.

 Tip

Generally, when define a new quantity implicitly (i.e. without giving an *explicit* formula), we should ask ourselves whether a *unique* quantity satisfies the definition. For example, a subset of $\mathbb{R}$ may have more than one upper bound. That a subset has only one *least* upper bound (if it has one at all) is something we need to prove.

Theorem 3.6. *If a subset of $\mathbb{R}$ has a least upper bound, then it is unique.*

Proof. Suppose u, v are both least upper bounds for a subset $S \subseteq \mathbb{R}$.¹ Then u is an upper bound for S , so since v is a *least* upper bound, $v \leq u$. Also, v is an upper bound for S , so since u is the *least* upper bound, $u \leq v$. Therefore $u = v$. $\square$

To prove a quantity is the least upper bound of a set, generally, one needs to show two things: prove that it's an upper bound, and then prove that any other upper bound is less than or equal to it.

Example 3.7. Here we'll do a careful proof that 3 is the supremum of $S = (-\infty, 3)$.

First, note that by definition S is all numbers less than 3, so for any $x \in S$, $3 > x$. Therefore 3 is an upper bound on S .

By way of contradiction, suppose $b < 3$ and b is also an upper bound on S .² But then consider $x = \frac{b+3}{2}$ so that $b < x < 3$. Since $x < 3$, we know $x \in S$. But $b < x$ which contradicts b being an upper bound on S .

¹I am using one of the standard proof skeleta for proving uniqueness. See Section ??.

²In Exercise ?? you were to show that something like this is the negation of 3 being the supremum. In logical symbols, saying 3 is the least upper bound is $(\forall \text{ upper bounds } b)(3 \leq b)$. And thus its negation is $(\exists \text{ an upper bound } b)(b < 3)$.

Note that doing a proof by contradiction above was not really necessary. Another option would be to say “suppose $b < 3$ ” and then prove b is not an upper bound.

Next we have some basic consequences of Theorem ???. The next theorem and its corollary are both often referred to as the “Archimedean Property” of $\mathbb{R}$.

Theorem 3.8 (Archimedean Property - Version 1). *For any $x \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $n > x$.*

Proof. By way of contradiction, suppose _____, which is the same as saying that x is an upper bound on $\mathbb{N}$. Then since $\mathbb{N}$ is _____ and bounded above, $\mathbb{N}$ has a least upper bound u . But then $u - \frac{1}{2} < u$, so $u - \frac{1}{2}$ is not an upper bound. Therefore there exists $n \in \mathbb{N}$ such that _____. But then

$$n + 1 > u - \frac{1}{2} + 1 = u + \frac{1}{2} > u,$$

and since $n + 1 \in \mathbb{N}$, we have contradicted _____. □

Exercise 3.9. Fill in the blanks for the proof above. For the first blank, you’ll need to negate the original statement.

Corollary 3.10 (Archimedean Property - Version 2). *For any $\epsilon > 0$, there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < \epsilon$.*

Exercise 3.11. Prove Corollary ???.

 Hint

Remember that when we call a theorem a “Corollary”, it means that you should use the preceeding theorem to prove it, and the proof will generally end up being quite short.

Exercise 3.12. Give a careful proof that 1 is the supremum of $S = \left\{ \frac{n}{n+1} : n \in \mathbb{N} \right\}$. Give a proof that is not a proof by contradiction.

 Hint

The same trick of taking an average like in Example ?? won’t necessarily work since that average might not be in S . Corollary ?? might be helpful here instead.

Exercise 3.13. Suppose $S \subseteq \mathbb{R}$ is a nonempty set that is bounded above. Let $c > 0$. We define the set

$$T = \{cx : x \in S\}.$$

Prove that $c \cdot \sup S$ is a least upper bound for T .

Exercise 3.14. Suppose S and T are nonempty subsets of $\mathbb{R}$ that are bounded above. What is $\sup(S \cup T)$? Prove your answer.

Exercise 3.15. Prove the Archimedean Property is true when the universe is $\mathbb{Q}$. This means you'll need to refrain from making any references to $\mathbb{R}$, and you of course can't use the Least Upper Bound Axiom.

In Part 2e of Exercise ?? and the follow-up discussion, we claimed that the existence of suprema would “plug holes” in the real numbers. In particular, the supremum of $S = \{x : x^2 \leq 3\}$, which we are assuming exists because of the Least Upper Bound Axiom, seems like it ought to be a number whose square is 3. We'll prove a more general version of this fact below.

Theorem 3.16. *For any $a \geq 0$, there exists a real number $x \geq 0$ such that $x^2 = a$.*

Proof Outline. Suppose $a \geq 0$. Define

$$S = \{y : y^2 \leq a\}$$

We will show that S is nonempty and bounded above so that we can claim $\sup S$ exists.

First, we note that $0^2 = 0 < a$, so $0 \in S$, and thus S is nonempty.

Next, I claim that $a + 1$ is an upper bound on S . This is true because

$$(a + 1)^2 = a^2 + 2a + 1 > a$$

So if $y \in S$,

$$y^2 \leq a < (a + 1)^2$$

so $y < a + 1$. Thus $a + 1$ is an upper bound on S , so S is bounded above, and $x = \sup S$ exists.

Now we must show $x^2 = a$. This is the hard part, and it boils down to showing that if $x^2 \neq a$, there must exist some z so that z^2 is strictly between x^2 and a , which we can use to lead to a contradiction. For example, if $x^2 < z^2 < a$, then $x < z$ and $z \in S$, which contradicts x being an upper bound on S . And if $a < z^2 < x^2$, then $z < x$, and for any $y \in S$, $y^2 < a < z^2$, so $y < z$; this would show that z is an upper bound less than x , contradicting x being the *least* upper bound.

All that remains is to show that such a z exists, which we'll do here, although the specifics get a bit technical.

(Finish later)

□

3.4 Review Questions

1. Give definitions of upper bound, least upper bound, and greatest element.
2. What does the Least Upper Bound Axiom say? What two things do we need to know about a set S in order to apply the Least Upper Bound Axiom?
3. For examples like those in Part 2 of Exercise ??, can you quickly identify the least upper bound when there is one?
4. What do the two versions of the Archimedean Property tell us?

4 Consequences of the Least Upper Bound Axiom

Highlights

- We define the greatest lower bound (also called the infimum) of a set and prove that these exist in $\mathbb{R}$ whenever a nonempty subset is bounded below.
- We prove the Nested Interval Theorem, which shows that the intersection of a nested sequence of closed intervals is nonempty.
- We show how the Nested Interval Theorem can be used to prove that $\mathbb{R}$ is uncountable.

Preparation

We start with a definition that should seem very natural given what we did in the last section.

Definition 4.1. Let $S \subseteq \mathbb{R}$ be a subset of the real numbers.

1. We say $b \in \mathbb{R}$ is a *lower bound* of S if for all $x \in S$, $b \leq x$. If S has a lower bound, we say S is *bounded below*.
2. We say that $l \in \mathbb{R}$ the *greatest lower bound* or *infimum* of S if for all lower bounds b of S , $l \geq b$, and in this case we write

$$l = \inf S.$$

3. If l is the greatest lower bound S and $l \in S$, we say that l is the *least element* of S .

Exercise 4.2. Find the infimum and least element of each set below or say why it doesn't have one.

1. $[0, \infty)$
2. $(-\infty, 0]$
3. $(0, 3]$
4. The set of upper bounds on $[0, 3]$.
5. The set of upper bounds on $[0, 3)$.

Parts 4 and 5 of Exercise ?? are meant to demonstrate another way of thinking about the Least Upper Bound Axiom: it tells us that sets of upper bounds *always* have a least element when they are non-empty.

Rather than make an assumption about infima existing, as we did with suprema, we can prove this as a theorem from the Least Upper Bound Axiom using this idea.

Theorem 4.3. *If a nonempty subset $S \subseteq \mathbb{R}$ has a lower bound, then S has a greatest lower bound in $\mathbb{R}$.*

Proof. Suppose $S \subseteq \mathbb{R}$ is nonempty and bounded below. Then let T be the set of lower bounds of S . Note that T is nonempty (since S was bounded below) and bounded above (by any element of S). Thus T has a least upper bound $l = \sup T$.

I claim that $l = \inf(S)$. First, we note that l is in fact greater than any other lower bound of S , since l is an upper bound on T . Also, since every element of S is an upper bound on T , we have $l \leq s$ for all $s \in S$ since l is the *least* upper bound of T . $\square$

Exercise 4.4. Try proving Theorem ?? another way. Consider the set $U = \{x : -x \in S\}$, and show that $-\sup U = \inf S$.¹

One more concept we'll need in order to get started on this section is the idea of taking an infinite union or intersection, which will be important for understanding the Nested Interval Theorem.

Exercise 4.5. For each example below, (i) draw the sequence of intervals on a number line, (ii) decide whether the sequence of intervals satisfies the conditions of Theorem ?? below, and (iii) identify the intersection $\bigcap_{i=1}^{\infty} [a_i, b_i]$.²

1. $[a_i, b_i] = [-\frac{1}{i}, \frac{1}{i}]$ for $i \in \mathbb{N}$
2. $[a_i, b_i] = [-i, i]$ for $i \in \mathbb{N}$
3. $[a_i, b_i] = [0, 1 + \frac{1}{i}]$ for $i \in \mathbb{N}$

¹We will use this kind of technique frequently to obtain the second two cases of a theorem we want to prove (e.g. one for increasing functions and one for decreasing function, or one for sups and one for infs).

²If you are rusty on unions and intersections, see Section ??.

4.1 Nested Interval Theorem

One distinguishing feature of the real numbers is, intuitively, if you keep zooming in on the number line, you are always zooming in *to a number*.

As a thought experiment, consider the following collection of intervals:

$$\begin{aligned} U_1 &= [3, 4] \\ U_2 &= [3.1, 3.2] \\ U_3 &= [3.14, 3.15] \\ U_4 &= [3.141, 3.142] \\ U_5 &= [3.1415, 3.1416] \\ U_6 &= [3.14159, 3.14160] \\ &\vdots \end{aligned}$$

Specifically, U_i is the closed interval between π rounded up and down to i significant figures.

If the universe is $\mathbb{R}$, there is a number that will be in U_i for all i : π . By contrast, in $\mathbb{Q}$, there is no number inside all of these sets—we’d be zooming in onto something that isn’t there. The Nested Interval Theorem below generalizes this idea.

Theorem 4.6 (Nested Interval Theorem). *Suppose a_i and b_i are real numbers with $a_i \leq b_i$ for all $i \in \mathbb{N}$ such that if we set $U_i = [a_i, b_i]$, we have $U_{i+1} \subseteq U_i$ for all $i \in \mathbb{N}$. In this case*

$$\bigcap_{i=1}^{\infty} U_i \neq \emptyset.$$

Setup Only. Assuming the hypotheses of the theorem, we note that since $U_{i+1} \subseteq U_i$, we have for all $i \in \mathbb{N}$

$$a_i \leq a_{i+1} \quad b_{i+1} \leq b_i.$$

In fact, from this, one can prove³ that $a_i \leq b_j$ for any $i, j \in \mathbb{N}$.

...

□

Exercise 4.7. Finish the proof of the Nested Interval Theorem.

The big hint here is that Nested Interval Theorem is true for $\mathbb{R}$ but not for $\mathbb{Q}$, so it must use the Least Upper Bound Axiom in some essential way.

³This is mathspeak for “I don’t feel like writing down all the details.” In this case, the claim is hopefully very believable, but the simple version of the proof requires some theorems about sequences we’ll prove later on in Chapter ??.

 Hint

To prove a set is nonempty, you should exhibit an element that lives in the set, *and then prove that element lives in the set.*

4.2 Uncountability of $\mathbb{R}$

Recall that a set S is called *countable* if there exists a surjective function $\mathbb{N} \rightarrow S$. Intuitively, it means that there is some way of listing the elements of S (possibly with repetition) so that every element of S eventually appears on the list.

It is clear that $\mathbb{N}$ is countable, but somewhat less obvious are the facts that $\mathbb{Z}$ and $\mathbb{Q}$ are countable⁴. However, $\mathbb{R}$ is uncountable; it has a strictly larger cardinality than $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$. Intuitively, the number of real numbers is a “higher level of infinity” than the number of natural/integer/rational numbers.

There are a few ways to prove that $\mathbb{R}$ is uncountable, but as we should now expect, all of them rely on the Least Upper Bound Axiom since uncountability is a property $\mathbb{R}$ has but $\mathbb{Q}$ does not. We will take the approach of using the Nested Interval Theorem, which in turn relies on the Least Upper Bound Axiom.

Theorem 4.8. $\mathbb{R}$ is uncountable.

Proof. By way of contradiction, suppose $\mathbb{R}$ is countable. Then there exists a surjective function $f : \mathbb{N} \rightarrow \mathbb{R}$. We will use this function to construct a sequence of nonempty nested closed intervals $\{U_i\}$ and apply the Nested Interval Theorem to arrive at a contradiction. The idea will be to construct U_{i+1} so that it excludes $f(i)$, and that way no value of f will lie in all of the intervals.

Let $a_1 = 0$ and $b_1 = 1$. For each $i \geq 1$, we recursively construct a_{i+1} and b_{i+1} using the following procedure:

- If $f(i) \notin [a_i, b_i]$, we let $a_{i+1} = a_i$ and $b_{i+1} = b_i$ (i.e. the interval remains unchanged).
- If $f(i) \in [a_i, \frac{b_i+a_i}{2}]$ (that is, $f(i)$ is in the left half of the interval), let $a_{i+1} = \frac{2b_i+a_i}{3}$ and $b_{i+1} = b_i$ (so that $[a_{i+1}, b_{i+1}]$ is the right third of the interval).
- If $f(i) \in (\frac{b_i+a_i}{2}, b_i]$ (that is, $f(i)$ is in the right half of the interval), let $a_{i+1} = a_i$ and $b_{i+1} = \frac{b_i+2a_i}{3}$ (so that $[a_{i+1}, b_{i+1}]$ is the left third of the interval).

⁴See Appendix ?? for a quick overview of why this is true.

Note that in each of the three cases $f(i) \notin [a_{i+1}, b_{i+1}]$. Letting $U_i = [a_i, b_i]$ for $i \geq 1$, this means that $f(i) \notin U_{i+1}$, and therefore no value $f(n)$ is in *all* of the closed intervals. That is, for all $n \in \mathbb{N}$,

$$f(n) \notin \bigcap_{i=1}^{\infty} U_i$$

However, by our assumption, every real number is equal to $f(n)$ for some $n \in \mathbb{N}$, so no real numbers are in the intersection, contradicting the Nested Interval Theorem. $\square$

We mention here that the proof above actually shows that $[0, 1]$ is uncountable, and can be adapted to prove the following.

Theorem 4.9. *Any interval $[a, b] \subseteq \mathbb{R}$ with $a < b$ is uncountable.*

Review Questions

1. Give the definitions of lower bound, bounded below, infimum, and least element.
2. What does the Nested Interval Theorem say? Give an example that shows how it distinguishes $\mathbb{R}$ from $\mathbb{Q}$.
3. What does it mean when we say $\mathbb{R}$ is uncountable?
4. How did we know we would need to use the Least Upper Bound Axiom or one of its consequences when proving $\mathbb{R}$ is uncountable?

Part II

Metric Spaces

5 Metric Spaces and Distance Functions

Highlights

- The definitions of “metric space” and “distance function”
- The usual distance functions on $\mathbb{R}^n$, including $\mathbb{R} = \mathbb{R}^1$.
- Properties of absolute values

5.1 Preparation

The limits that you learn about in calculus are often described in terms of a variable getting “closer and closer” or “arbitrarily close” to some quantity. In order to say more precisely what it means to be “close”, though, we need to be precise about what we mean by “distance” in the first place.

In this chapter we’ll talk about a fundamental construction in the theory of real analysis, which is the concept of a metric space. Basically, a metric space will be any set with a reasonable (and we’ll make precise what we mean by “reasonable”) notion of distance.

You are likely familiar with the distance formulas in $\mathbb{R}^2$ and $\mathbb{R}^3$, and we will use these two spaces as our mental model for how to think about metric spaces. We’ll see that distance can reasonably be defined in other contexts as well, including in the real number system.

For the moment, play around with the applet below to see if you can think about what formula would give us the distance between the real numbers a and b .

Our philosophy will be that we will:

- Gain intuition from $\mathbb{R}^2$ and $\mathbb{R}^3$,
- Write our proofs for any metric space, and
- Apply what we learn to $\mathbb{R}$.

For whatever reason, it’s easy for our brains to think about geometry in two and three dimensions than it is in one dimension. And proving theorems about general metric spaces is

sometimes easier than proving the corresponding theorems in $\mathbb{R}$ since there are fewer tools to keep track of¹.

As a side benefit to this approach, the theory of metric spaces is useful more broadly in pure and applied math since we often want to think about how to define the “distance” between objects in a reasonable way. As a teaser of this concept, what is a reasonable way to define the distance between two *functions*?

As we said earlier, we will define a metric space to be any set in which we can reasonably define distance. We begin by making the idea of “reasonableness” precise here.

Definition 5.1. A **metric space** is a nonempty set X equipped with a function $d : X \times X \rightarrow \mathbb{R}$ called a **distance function** with the following properties.

1. (Positivity) For any $a, b \in X$, $d(a, b) \geq 0$.
2. (Positive Definiteness) For any $a, b \in X$, $d(a, b) = 0$ if and only if $a = b$.
3. (Symmetry) For any $a, b \in X$, $d(a, b) = d(b, a)$.
4. (Triangle Inequality) For any $a, b, c \in X$, $d(a, b) + d(b, c) \geq d(a, c)$.

Exercise 5.2. Explain why the usual distance formula in $\mathbb{R}^2$,

$$d((x_1, y_1), (x_2, y_2)) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2},$$

satisfies the properties in Definition ???. Give a proof for Positivity, Positive-Definiteness, and Symmetry, and provide a picture with an intuitive explanation for the triangle inequality.

While we are used to measuring distances in two and three dimensions, we can define distances in other contexts as well.

Exercise 5.3. Let S be any set. Prove that $d : S \times S \rightarrow \mathbb{R}$ defined for any $x, y \in S$ by

$$d(x, y) = \begin{cases} 0 & x = y \\ 1 & x \neq y \end{cases}$$

is a distance function for S (i.e. that it satisfies the four properties). We call this the **trivial** metric for S .

¹For instance, most metric spaces don’t have arithmetic or inequalities, so we won’t be distracted by these features of $\mathbb{R}$ in contexts where they aren’t relevant.

To talk about a metric space, we really ought to specify both the set X and the distance function d . For example, it's technically possible to equip $\mathbb{R}^2$ with the trivial metric from Exercise ?? (although I'm hard-pressed to think of why this would be useful). A somewhat more practical alternative metric on $\mathbb{R}^2$ is the subject of Exercise ??, but we'll refer to the distance function from Exercise ?? as the “standard” distance function for $\mathbb{R}^2$.

Exercise 5.4. Make a guess about how we will define the “standard” distance function that will turn $\mathbb{R}$ into a metric space.

5.2 Distances in $\mathbb{R}^n$

You are likely familiar with the distance formulas in $\mathbb{R}^2$ and $\mathbb{R}^3$. Generalizing those formulas, the standard distance function in $\mathbb{R}^n$ is given by

$$d((a_1, \dots, a_n), (b_1, \dots, b_n)) = \sqrt{(b_1 - a_1)^2 + \dots + (b_n - a_n)^2}. \quad (5.1)$$

The proofs that this distance function satisfies Positivity, Positive-Definiteness, and Symmetry are straightforward generalizations of what you did in Exercise ??. The proof that the standard distance function satisfies the Triangle Inequality is much more involved, and we defer it to Theorem ?? in Appendix ??.

More important to us at the moment, though, is how to define distances in $\mathbb{R}$. If we apply the formula (??) to the case where $n = 1$ we get² $d(a, b) = \sqrt{(b - a)^2}$. This ends up simplifying to $|b - a|$, which hopefully matches your guess in Exercise ??.

Thus we define the **standard distance function on $\mathbb{R}$** to be

$$d(a, b) = |b - a|.$$

From now until the end of your days, whenever you see an absolute value—especially the absolute value of a difference—you should think “distance”.

Now that we see absolute values will be our main tool for talking about distances in the real numbers, it will be worthwhile to review some key properties of absolute values.

²Here we set $a = (a_1)$, and $b = (b_1)$ to simplify the notation.

5.3 Absolute Values

We can define the absolute value function for any $x \in \mathbb{R}$ via

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

Make sure you understand how the second case matches what you know about absolute values before moving on!

Theorem 5.5. Suppose $a, b \in \mathbb{R}$.

1. $|a| \geq 0$.
2. $|-a| = |a|$.
3. $|a| = \max\{a, -a\}$.
4. $|a| \geq a$.
5. $|ab| = |a||b|$.
6. $|a + b| \leq |a| + |b|$.
7. $|a - b| \geq ||a| - |b||$.

Exercise 5.6. Prove Theorem 5.5. Note that since the definition of absolute value is piecewise, you may need to do case-by-case proofs here.

Exercise 5.7. Prove that the standard distance function $d(x, y) = |y - x|$ for $\mathbb{R}$ satisfies the Triangle Inequality.

Exercise 5.8. Let X be a metric space with distance function d and let $x, y \in X$. Suppose that for all $\epsilon > 0$, $d(x, y) \leq \epsilon$. Prove that $x = y$.

Tip

In this section, you should cite Positivity, Positive-Definiteness, and the Triangle Inequalities whenever you use them (don't bother with Symmetry). After this section, you need only cite the Triangle Inequality.

Exercise 5.9. Let X be a metric space with distance function d . Prove that for any $a, b, c \in X$,

$$|d(a, b) - d(b, c)| \leq d(a, c)$$

Exercise 5.10. Prove the “polygon inequality”, which says that for any $x_1, \dots, x_n$ with $n \geq 3$ in a metric space X with distance function d ,

$$d(x_1, x_2) + d(x_2, x_3) + \cdots + d(x_{n-1}, x_n) \geq d(x_1, x_n)$$

Hint: this is a good candidate for mathematical induction. It’s also a good example of how drawing a picture in two dimensions can help give you intuition about what happens in any metric space.

Exercise 5.11. Here we will consider $\mathbb{R}^2$ with a different distance function. Define

$$d_1((x_1, y_1), (x_2, y_2)) = |y_1 - x_1| + |y_2 - x_2|$$

Prove that d_1 does in fact satisfy the four conditions of being a metric space.

Exercise 5.12. Here we will consider $\mathbb{R}$ with a different purported distance function. Define

$$d_{\text{inv}}(a, b) = \begin{cases} \frac{1}{|a-b|} & a \neq b \\ 0 & a = b \end{cases}$$

Show that d_{inv} is *not* a distance function.

Exercise 5.13. If X is a metric space with distance function d , and $S \subseteq X$, then we can think of S itself as a metric space using the same distance function restricted to S . In this case we say S is a **subspace** of X .

1. If we treat $\mathbb{Q}$ as a subspace of $\mathbb{R}$, prove that the distance between any two points in $\mathbb{Q}$ is a rational number.
2. If we treat $\mathbb{Z}^2$ (the set of points in $\mathbb{R}^2$ both of whose entries are integers), as a subspace of $\mathbb{R}^2$, is the distance between two points in $\mathbb{Z}^2$ always an integer? Explain.

Review Questions

1. State the four conditions a distance function on a metric space must satisfy.
2. What is the standard distance function in $\mathbb{R}^n$? What about in $\mathbb{R}$?
3. Which properties of absolute value from Theorem ?? did you already know? Which were new to you?

6 $\mathbb{R}$ as a Metric Space

Highlights

- See how distance interacts with previously established properties of $\mathbb{R}$ (arithmetic, inequalities, and least upper bounds).
- Equivalent conditions to being a least upper bound.
- Use distance to investigate the relationships among the integers, rationals, irrationals, and reals.

6.1 Preparation

In this section, we will focus on thinking about the real number system as a metric space. We start with what will end up being one of the most-cited theorems in the course.

Theorem 6.1. *Let $r > 0$ and $a \in \mathbb{R}$. Then $|a - x| < r$ if and only if $x \in (a - r, a + r)$.*

Exercise 6.2.

1. Draw a picture of what this Theorem ?? is saying on the number line.
2. Prove Theorem ??.
3. Rewrite $|a - x| < r$ as a statement about distance in $\mathbb{R}$.
4. Rewrite $x \in (a - r, a + r)$ as an inequality.

Theorem ?? is foundational because it connects a statement about analysis (distances and closeness) to a statement about geometry (subsets of $\mathbb{R}$). In general, mathematicians love “if and only if” statements because they often let you translate a statement in one context to a statement in another, where you may have a whole new set of tools you can apply. Together with Parts 3 and 4 from Exercise ??, you now have four toolbags you can use to analyze the statement: absolute values, intervals, distances, and inequalities.

Indeed, the connections furnished by Theorem ?? can be used to help us prove other theorems relating the concepts we’ve encountered so far.

Theorem 6.3. *Suppose b is an upper bound on a nonempty subset $S \subseteq \mathbb{R}$. Then the following are equivalent.*

1. $b = \sup S$
2. For all $\epsilon > 0$ there exists $x \in S$ such that $x \in (b - \epsilon, b]$.
3. For all $\epsilon > 0$ there exists $x \in S$ such that $|b - x| < \epsilon$.

Proof. (Include link to my video proof?) □

Note

When we encounter a new concept, we will often have a TFAE (“the following are equivalent”) theorem that follows giving us alternative ways of understanding the new concept. Since these theorems are so useful, I will color-code them with a lightseagreen vertical bar.

Exercise 6.4. Carefully prove that 5 is the supremum of $[0, 5)$ using Part 2 or 3 of Theorem ??.

Exercise 6.5. Write down a “the following are equivalent” theorem about infima analogous to Theorem ?? and prove it.

6.2 The max and min of a Finite Set

In Theorem ??, I used the notation $\max\{a, -a\}$ to mean whichever of these two numbers is greater. Here we will make that notation precise with some words of caution about when you can use it.

Theorem 6.6. Suppose $F \subseteq \mathbb{R}$ is a nonempty finite set. Prove that $\sup F$ exists and $\sup F \in F$ (i.e. that F has a greatest element). In this case we write $\max F$ for the greatest element of F .

For example, $\max\{3, 7, 5\} = 7$.

Proof. Let n be the number of elements of F . We will prove that F has a greatest element by induction on n .

As a base case, suppose $n = 1$. Then $F = \{x_1\}$. In this case $x_1 \geq x_1$, so x_1 is an upper bound on F . And any upper bound b on F must have $b \geq x_1$, so x_1 is the least upper bound. And of course $x_1 \in F$.

Now suppose that $k \in \mathbb{N}$ and any set with k elements has a greatest element. Now suppose $F = \{x_1, \dots, x_k, x_{k+1}\}$ has $k + 1$ elements. Define $E = \{x_1, \dots, x_k\}$ so that E has k elements and therefore a greatest element g . If $x_{k+1} \geq g$, then let $b = x_{k+1}$, and otherwise, let $b = g$. Either way, $b \geq x_{k+1}$ and $b \geq g \geq x_i$ for all $i = 1, \dots, k$ (so b is an upper bound), and either way $b \in F$. And if u is any upper bound on F , then $u \geq b$ and $u \geq x_{k+1}$, so $u \geq b$. Thus, b is the *least* upper bound of F , and since $b \in F$, b is the greatest element. □